LECTURE 2

Notation

The following notation is equivalent:

(1) $2, 4, 8, 16,$

(2) $\{2^n\}$

(3) $\{2^n\}_{n=2}^{\infty} = 4, 8, 16, 32, \ldots$

(4) $a_n = 2^n$

Ex

Write the first four terms of the sequence.

$\frac{2n-1}{n!}$ $\quad 1, \frac{3}{2}, \frac{5}{6}, \frac{7}{24}$

Ex

Find a formula for the nth term of the sequence.

$\frac{3}{4}, \frac{5}{8}, \frac{7}{16}, \frac{9}{32}, \ldots \qquad a_n = \frac{2n+1}{2^{n+1}}$

$-\frac{1}{3}, \frac{7}{9}, \frac{-25}{27}, \frac{79}{81}, \ldots \qquad a_n = (-1)^n \cdot \frac{3n-2}{3^n}$

$1, -\frac{1}{2}, \frac{1}{6}, -\frac{1}{24}, \ldots \qquad a_n = (-1)^{n+1} \cdot \frac{1}{n!}$

Def'n

Let $\{a_n\}$ be a sequence. The limit of the sequence is defined as $\lim_{n \to \infty} a_n$ if it exists.

If the limit exists, $\{a_n\}$ <u>converges</u>.
If the limit DNE, $\{a_n\}$ <u>diverges</u>.

Theorem

Let $f(x)$ be a function such that $f(n) = a_n$ for all $n \in \mathbb{N}$. If $\lim_{x \to \infty} f(x) = L$, then

$$\lim_{n \to \infty} a_n = L.$$

Sequences, convergence, divergence, equivalence of limits of functions and limits of sequences

CONTINUED

Ex Determine if the sequence converges or diverges.

$$\left\{ \frac{n^3 + n}{2 + 5n^3} \right\} = \lim_{n \to \infty} \frac{n^3 + n}{2 + 5n^3}$$

$$= \lim_{n \to \infty} \frac{n^3/n^3 + n/n^3}{2/n^3 + 5n^3/n^3}$$

$$= \lim_{n \to \infty} \frac{1 + n/n^3}{2/n^3 + 5}$$

$$= \frac{1 + 0}{0 + 5}$$

$$= \frac{1}{5}$$

Hence it converges.

$$\left\{ \frac{e^n}{n + 1} \right\} = \lim_{n \to \infty} \frac{e^n}{n + 1}$$

$$\overset{H}{=} \lim_{n \to \infty} e^n$$

$$= \infty$$

Hence it diverges.

Def'n Let $\{a_n\}$ be a sequence such that there exists b_n with $b_n > 0$ and $a_n = (-1)^n b_n$ or $a_n = (-1)^{n+1} b_n$.

$\{a_n\}$ is called an <u>alternating sequence</u>.

Ex Show $\left\{ \frac{(-1)^n n^2}{2^n} \right\}$ is alternating by finding b_n.

$$b_n = \frac{n^2}{2^n} > 0$$

Theorem If $\lim_{n \to \infty} |a_n| = 0$, then $\lim_{n \to \infty} a_n = 0$

alternating sequences, absolute value convergence to zero

CONTINUED

Ex

Show $\left\{ \frac{(-1)^{n+1}}{n} \right\}$ converges to 0.

Let $a_n = \frac{(-1)^{n+1}}{n}$. We have

$$\lim_{n \to \infty} \left| \frac{(-1)^{n+1}}{n} \right| = \lim_{n \to \infty} \frac{1}{n} = 0$$

By theorem, $\lim_{n \to \infty} \frac{(-1)^{n+1}}{n} = 0$.